VI. (25 points) In Fig. A, two thin parallel plates are ideal conductors, spaced *d*, infinite in *y*. An EM wave between them travels normal to *y*, along positive *x* as a travelling wave， with electric field

$E = E_0 \sin(2\pi z / \lambda_z)\sin(2\pi x / \lambda_x - \omega t)$. ω is

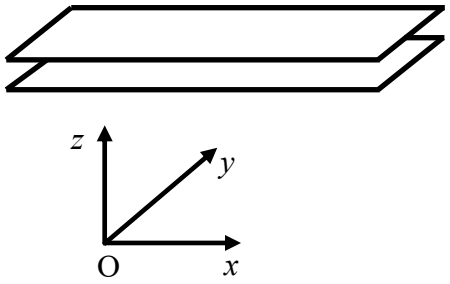


Fig. A

angular frequency; t is time, λ_z, λ_x are unknown.

Combining these makes practical microwave transmitting antennas, replacing huge conventional parabolic dishes and greatly reducing antenna cost.

1. Prove that λ_z can take only these values:

$$\lambda_z = 2d / m, \quad m = 1, 2, 3\ldots$$

2. For *m*=1, find λ_x .

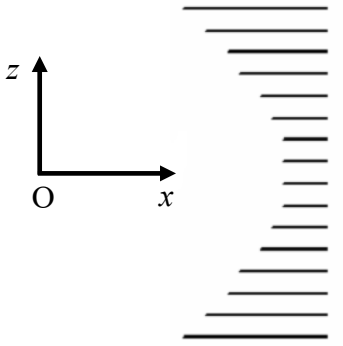


Fig. B

3. Stack equal-gap plates of unequal lengths, parallel and symmetric about an infinite *y* line source at O. Their right ends align; their lengths vary.
Fig. B is the cross-section in the page. At the right end, a plane EM wave is to emerge along *x*. Find the equation of the left boundary in the *x*O*z* section (Fig. B). (Take *m*=1; the horizontal distance from the source to the stack's left end is *L*）。